Module 1: Getting Started with Artificial Intelligence — From Myths to Machines

Welcome to the World of Artificial Intelligence (AI)!

> "AI is not science fiction anymore — it's science fact."

From robot sidekicks in Hollywood blockbusters to the invisible intelligence that recommends your next favorite show on Netflix — Artificial Intelligence (AI) is everywhere. But what exactly is AI? And how did it go from being a myth in ancient Greek stories to driving autonomous vehicles on real streets? You're about to find out.

In this module: Introduction · Overview and Objectives · Course Materials

Introduction

What Is Artificial Intelligence?

Let's get this straight right away:

- AI is not just about robots with red eyes taking over the world.
- AI is also not magic.
- AI is simply the ability of a computer or machine to mimic human intelligence:
  - learn from data;
  - solve problems;
  - make decisions; and
  - even learn from mistakes (just like you do!).

AI in Pop Culture: Fun, But Fictional

Before diving deep, let's debunk some fun AI myths:

Myths	Reality
AI will become self-aware and rule the world.	Nope. AI today is task-specific and lacks emotions or goals.
AI can think like humans.	Not quite. AI processes data, but it doesn't "feel" or "think" like you.
Robots = AI.	Not always. Robots can exist without AI, and AI can exist without robots.

Quick Activity

Think of a movie, show, or book with AI in it. Was it realistic? Why or why not? Share your thoughts with your instructor and peers!

AI Is Already Around You

Believe it or not, AI is already a part of your daily life. Here are just a few examples:

Everyday Activity	AI Involved
Asking Siri or Alexa	Natural Language Processing
Using Google Maps	Route optimization with Machine Learning
Streaming on Spotify	Music recommendations using Deep Learning
Seeing a Tesla drive itself	Computer Vision + AI decision-making

So yes, every time you say "Hey Google" — you're using AI.

A Quick Peek into History

AI didn't just show up overnight. Here are some cool highlights:

- **Ancient Greece:** The myth of **Talos**, a bronze automaton guarding Crete. View source PDF on Canvas · Download source PDF
- **1956:** The term "Artificial Intelligence" was coined at Dartmouth College. View source PDF on Canvas · Download source PDF
- **1997:** IBM's **Deep Blue** defeated world chess champion Garry Kasparov.
- **Now:** The rise of **Deep Learning** revolutionized how AI interprets images, speech, and more. View source PDF on Canvas · Download source PDF

Why AI Matters to STEM Students Like You

No matter what you study — biology, physics, chemistry, engineering, or mathematics — AI is reshaping your field:

- Biologists use AI to detect diseases earlier.
- Chemists simulate molecules with AI before stepping into a lab.
- Engineers design smarter, safer cities using AI-based data analysis.
- Mathematicians model AI algorithms for faster computation.

Whether you're heading into medicine or robotics, AI is a tool you'll want in your STEM toolbox.

What's next? Module 1: Overview and Objectives — What are we learning and why?

Overview and Objectives

Overview

This module introduces foundational concepts in Artificial Intelligence (AI) for STEM students without prior computer science experience. Students will:

- Understand what AI is and why it matters in science and society.
- Learn the history of AI, key milestones, and its influence on modern life.
- Differentiate between AI, Machine Learning (ML), and Deep Learning (DL).
- Grasp the role of intelligent agents and Artificial General Intelligence (AGI).
- Explore ethical implications of AI.

Learning Objectives

By the end of this module, you will be able to:

1. **Define** what Artificial Intelligence (AI) is and explain its role in modern society.
2. **Describe** the historical development of AI, from early ideas and myths to real-world technologies.

3. **Identify and explain** key terms and definitions used in AI, including algorithms, models, data, and intelligence.
4. **Differentiate** between Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) with real-world examples.
5. **Explain** what intelligent agents are and how they interact with environments to solve problems.
6. **Discuss** the concept of Artificial General Intelligence (AGI), how it differs from current AI systems, and its potential future uses.
7. **Evaluate** popular culture depictions of AI and compare them to real-world capabilities.
8. **Reflect on and discuss** ethical questions related to the development and use of AI in STEM fields.

What's next? Course Materials.

Course Materials

Readings

In the following readings, you'll learn:

1. **Introduction to Artificial Intelligence** — What AI really is, how it works, and why it matters in science and society. View on Canvas · Download PDF
2. **History of AI** — Key milestones and AI's influence on modern life. View on Canvas · Download PDF
3. **The Difference Between AI, ML, and DL.** View on Canvas · Download PDF
4. **What Makes an Intelligent Agent… Intelligent?** View on Canvas · Download PDF
5. **The Wild World of Artificial General Intelligence (AGI).** View on Canvas · Download PDF
6. **Exploring the Ethical Implications of AI.** View on Canvas · Download PDF
7. **Important Definitions When Learning Artificial Intelligence.** View on Canvas · Download PDF

PowerPoint Slides

Understanding_AI_PPT_Presentation.pptx — View on Canvas · Download PPTX

Internet-Accessible Lecture Videos

See *Module 1: Further Learning — Open Educational Resources*.

Google Colab

At the instructor's discretion.

Articles

- Towards Data Science
- AI4ALL

Further Reading

- Elements of AI — University of Helsinki's free AI course for beginners.

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